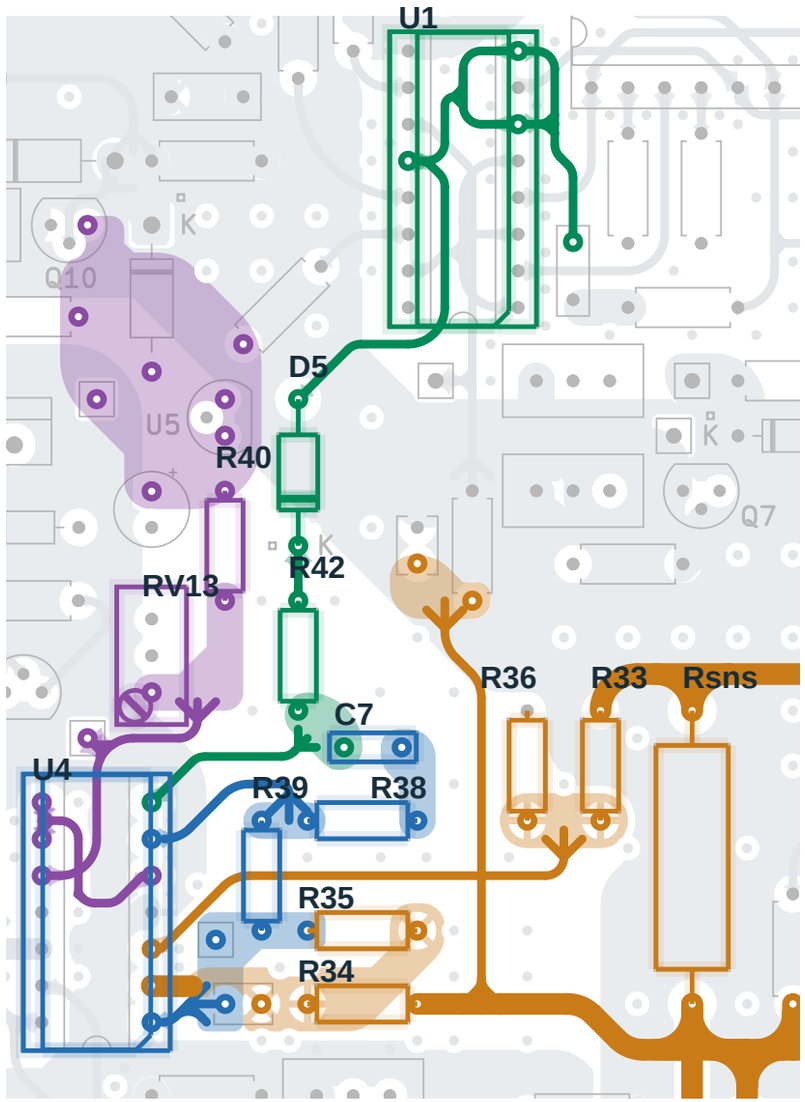
Current sensing and limiting

Actual KiCad copper + component highlights | Top-side projection of both outer layers



Differential sensing
R33 / R34 pick up opposite sides of Rsns. R35 / R36 complete the amplifier network.

Amplified signal
U4A output at pin 1 reaches U4B through R39. U4 contains several amplifier sections.

Adjustable limit
R40 / RV13 set the threshold; U4C buffers it for U4B.

Limiting action
U4B output reaches the error-amplifier node through R42 and D5.

A shared color may span series components. Their pads and copper breaks remain distinct.

Rsns → R33-R36 → U4A → R39 / U4B → R42 / D5 → EA node

Signal flow, not load-current flow. RV13 -> U4C -> U4B provides the adjustable comparison threshold.

Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.